SATGURU STUDY CENTRE
Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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CLASS : 9

COMMON ANNUAL EXAMINATION-2024-25

MATHEMATICS

Time Allowed : 3.00 Hours] **[Max. Marks : 100**

PART - I

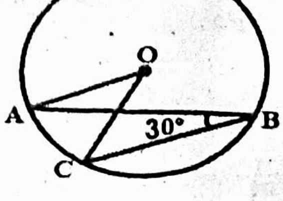
I. Choose the correct Answer. **14x1=14**

1. If $A \cup B = A \cap B$, then
   a) $A \neq B$ b) $A = B$ c) $A \subset B$ d) $B \subset A$
2. If $U = \{x : x \in N \text{ and } x<10\}$, $A = \{1, 2, 3, 5, 8\}$ and $B = \{2, 5, 6, 7, 9\}$ then $n[(A \cup B)']$ is ----------
   a) 1 b) 2 c) 3 d) 8
3. Which one of the following is an irrational number
   a) $\sqrt{25}$ b) $\sqrt{9/4}$ c) $7/11$ d) π
4. $(0.000729)^{-3/4} \times (0.09)^{-3/4} =$
   a) $\frac{10^3}{3^3}$ b) $\frac{10^5}{3^5}$ c) $\frac{10^2}{3^2}$ d) $\frac{10^6}{3^6}$
5. The root of the polynomial equation $2x+3=0$ is
   a) $\frac{1}{3}$ b) $\frac{-1}{3}$ c) $\frac{-3}{2}$ d) $\frac{-2}{3}$
6. Degree of the constant polynomial is --------------
   a) 3 b) 2 c) 1 d) 0
7. The interior angle made by the side in a parallelogram is 90° then the parallelogram is a
   a) Rhombus b) Rectangle c) Trapezium d) Kite
8. The angles of a triangle are $(3x-40)^\circ$, $(x+20)^\circ$ and $(2x-10)^\circ$ then the value of x is
   a) 40° b) 35° c) 50° d) 45°
9. If $(x+2, 4) = (5, y-2)$, then the coordinates (x,y) are
   a) (7,12) b) (6,3) c) (3,6) d) (2,1)
10. If Q_1, Q_2, Q_3, Q_4 are the quadrants in a cartesian plane then $Q_2 \cap Q_3$ is -----------
    a) $Q_1 \cup Q_2$ b) $Q_2 \cup Q_3$ c) Null set d) Negative x axis
11. The value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \ldots \tan 89^\circ$ is
    a) 0 b) 1 c) 2 d) $\frac{\sqrt{3}}{2}$
12. The lateral surface area of a cube of side 12cm is
    a) 144 cm² b) 196 cm² c) 576 cm² d) 664 cm²
13. The mean of the square of first 11 natural numbers is
    a) 26 b) 46 c) 48 d) 52
14. Which one of the following cannot be taken as probability of an event?
    a) 0 b) 0.5 c) 1 d) -1

PART - II

II. Answer any 10 questions. Question No. 28 is compulsory. **10x2=20**

15. If A = {6, 7, 8, 9} and B = {8, 10, 12} find $A \Delta B$.
16. If $A = \{a, \{a, b\}\}$, Write all the subsets of A.
17. Express $\frac{1}{13}$ in decimal form. Find the length of the period of decimals.
18. What should be added to $2x^3+6x^2-5x+8$ to get $3x^3-2x^2+6x+15$?
19. Expand $(m+5)(m-8)$.
20. If O is the centre of the circle and $\angle ABC = 30^\circ$ then find $\angle AOC$.



21. The centre of a circle is (-4, 2). If one and of the diameter of the circle is (-3, 7) then find the other end.
22. Find the distance between the points (-4, 3) , (2, -3).
23. Evaluate $\tan 60^\circ \cot 60^\circ$.
24. Find the TSA and LSA of the cube whose side is 5 cm.
25. A cubical milk tank can hold 125000 litres of milk. Find the length of its side in metres.
26. The arithmetic mean of 6 values is 45 and if each value is increased by 4, then find the arithmetic mean of new set of values.
27. The probability of guessing the correct answer to a certain question is $^{x}/_{3}$. If the probability of not guessing the correct answer is $^{x}/_{3}$, then find the value of x.
28. In a distribution, the mean and mode are 66 and 60 respectively. Calculate the median.

PART - III

III. Answer the following any 10 questions.Q.No.42 is compulsory. **10x5=50**

29. In a group of 100 students, 85 students speak Tamil, 40 students speak English, 20 students speak French, 32 speak Tamil and English, 13 speak English and French and 10 speak Tamil and French. If each student knows atleast any one of these languages, then, find the number of students who speak all these three languages.
30. Verify : $A - (B \cap C) = (A-B) \cup (A-C)$ using Venn diagrams.
31. Simplify $2\sqrt[3]{40} + 3\sqrt[3]{625} - 4\sqrt[3]{320}$
32. If $(x+a)(x+b)(x+c) = x^3 + 14x^2 + 59x + 70$, Find the value of

    i) $a + b + c$ ii) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ iii) $a^2 + b^2 + c^2$ iv) $\frac{a}{bc} + \frac{b}{ac} + \frac{c}{ab}$
33. Factorise $x^3 - 5x^2 - 2x + 24$.
34. Solve $2x - y = 3$; $3x + y = 7$.
35. Two circles of radii 5 cm and 3cm intersect at two points and the distance between their centres is 4 cm. Find the length of the common chord.
36. If $\sec\theta = \frac{13}{5}$, then show that $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta} = 3$
37. Find the area of an equilateral triangle whose perimeter is 180 cm.
38. The length, breadth and height of a cuboid are in the ratio 7 : 5 : 2, Its volume is 35840 cm^3. Find its dimensions.
39. Find the mean for the data.

CI	100 - 120	120- 140	140 - 160	160 - 180	180 - 200	200 - 220	220 - 240
Frequency	10	8	4	4	3	1	2

40. Two dice are rolled, find the probability that the sum is i) equal to 1 ii) equal to 4 iii) less than 13
41. Find the points of trisection of the line segment joining (-2, -1) and (4, 8)
42. The vertices of a triangle are (1, 2), (h, -3) abd (-4, k) If the centroid of the triangle is at the point (5, -1) then find the value of $\sqrt{(h+k)^2 + (h+3k)^2}$

PART - IV

IV. Answer all the questions. **2x8=16**

43. a) Construct the centroid of ΔPQR whose sides are PQ = 8 cm, QR = 6 cm, RP = 7 cm.

    (OR)

    b) Draw a triangle ABC, where AB = 8 cm, BC = 6 cm and $\angle B = 70^\circ$ and locate its circumcentre and draw the circumcircle.
44. a) Draw the graph for $y = 4x - 1$.

    (OR)

    b) Solve by graphically: $y = 2x + 1$; $-4x + 2y = 2$.